

Approximating Irrational Roots on a Number Line

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

- | | |
|---|--------------------------------------|
| 1. lower whole number = 5, upper whole number = 6 | 5. 8.49 |
| 2. 2.8 | 6. 12.2 ft |
| 3. 6.7 ft | 7. (a) to the nearest tenth = 6.3, — |
| 4. lower whole number = 10, upper whole number = 11 | 8. 9.75 |
| | 9. 9 ft |
| | 10. (a) the estimate = 7.07, — |
-

What is verified

8 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 7, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.NS.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

10. *If they got 25: halved the number instead of taking its square root*

1. $\sqrt{30}$ lies between which two whole numbers? Name the perfect squares used.

Solution: The perfect squares on either side of the radicand are 25 and 36, so $25 < 30 < 36$. Taking square roots across the inequality keeps the order, giving $5 < \sqrt{30} < 6$:
between

and

.

5

6

2. Give $\sqrt{8}$ to the nearest tenth.

Solution: $4 < 8 < 9$, so the value is between 2 and 3. Squeeze with tenths: $2.8^2 = 7.84$ and $2.9^2 = 8.41$, so the root lies between them and is nearer the lower one.

2.8

3. A square rug covers 45 square feet. Side to the nearest tenth of a foot?

Solution: For a square, area is side times side, so the side is the square root of the area. $36 < 45 < 49$ puts it between 6 and 7 feet; then $6.7^2 = 44.89$ and $6.8^2 = 46.24$ trap the area, and 44.89 is the closer of the two.

6.7 ft

4. $\sqrt{115}$ lies between which two whole numbers? Name the perfect squares used.

Solution: The perfect squares either side are 100 and 121, and $100 < 115 < 121$, so the root lies between

10

and

11

5. Give $\sqrt{72}$ to the nearest hundredth.

Solution: $64 < 72 < 81$ gives the bracket 8 to 9. Tenths: $8.4^2 = 70.56$ and $8.5^2 = 72.25$. Hundredths: $8.48^2 = 71.9104$ and $8.49^2 = 72.0801$, which trap the radicand.

8.49

6. A square vegetable bed covers 150 square feet. Side to the nearest tenth of a foot?

Solution: The side is the square root of the area. $144 < 150 < 169$ puts it between 12 and 13 feet; $12.2^2 = 148.84$ and $12.3^2 = 151.29$ trap the area, and the lower one is closer.

12.2 ft

7. Locate $\sqrt{40}$: (a) to the nearest tenth, (b) marked on the number line.

Solution (a): $36 < 40 < 49$ brackets the root between 6 and 7. Tenths: $6.3^2 = 39.69$ and $6.4^2 = 40.96$, so the root is just above the lower one.

6.3

Solution (b) — instructor-judged. The dot belongs between the ticks at three tenths and four tenths past the sixth whole number, close to the left one, and labelled with the radical or the rounded decimal. A dot outside that tenth-wide interval is not correct.

8. Give $\sqrt{95}$ to the nearest hundredth.

Solution: $81 < 95 < 100$ brackets it between 9 and 10. Tenths: $9.7^2 = 94.09$ and $9.8^2 = 96.04$. Hundredths: $9.74^2 = 94.8676$ and $9.75^2 = 95.0625$, which trap the radicand, and the upper one is nearer.

9.75

9. A square patch of ground has area 80 square feet, and tarps come only in whole numbers of feet on a side. Smallest tarp that covers it?

Solution: The patch's side is $\sqrt{80}$, and $64 < 80 < 100$ places that between 8 and 10; more tightly, $8^2 = 64$ is too small and $9^2 = 81$ is the first whole square at or above the area. An eight-foot tarp would leave part of the patch bare, so round *up*, not to the nearer whole number.

9 ft

10. **Challenge.** Ravi estimates $\sqrt{50} \approx 25$. (a) Give $\sqrt{50}$ to the nearest hundredth. (b) Explain his error.

Solution (a): $49 < 50 < 64$ brackets the root just above 7. Hundredths: $7.07^2 = 49.9849$ and $7.08^2 = 50.1264$ trap the radicand.

7.07

Solution (b) — instructor-judged. Ravi halved the radicand instead of looking for the number whose square is the radicand. The squaring check settles it: his value squared is 625, which is more than twelve times the number under the radical, while the correct estimate squared lands within a hundredth of it. A full answer names the halving error and shows that squaring check; the bracketing whole numbers are a fair sanity check to cite as well.